

Response to Reviewers

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Title: Texture Resonance Retrieval for Retrieval-Grounded Editable Audio Effect Preset Selection

Dear Editor and Reviewers,

We thank the reviewers for the detailed critiques. The central lesson from the reviews was that our earlier package mixed a verified retrieval benchmark with broader system claims that were not supported to the same standard. In the revision, we therefore prioritized *evidence hardening* over scope expansion. The revised manuscript now makes fewer claims, but each retained claim is tied to the verified P0 experiment package or a canonical repository artifact.

Canonical Evidence Sources

The revised submission uses the following repository artifacts as the sole experimental sources of truth:

- Experiments/tmm/protocolA_split_audit.md
- Experiments/tmm/protocolA_audio_grouped_leakage_report.json
- Experiments/AblationStudies/protocolA_audio_grouped_objective_stats.md
- Experiments/AblationStudies/protocolA_audio_grouped_per_query_metrics.csv
- Experiments/AblationStudies/outputs/p0_e2/e2_summary.csv
- Experiments/AblationStudies/outputs/p0_e2/e2_stats.json
- Experiments/AblationStudies/outputs/p0_e3_shared_e2_test/e3_overall_summary.csv
- Experiments/AblationStudies/outputs/p0_e3_shared_e2_test/e3_summary.csv
- Experiments/E5_Ablations/outputs/p0_near_dup/p0_near_dup_sensitivity.csv
- Experiments/E2_SOTABaselines/outputs/p0_mlp_regressor/mlp_regressor_results.json
- Experiments/mushura/results_report.md

We do not use the legacy retrieval report, the deprecated Protocol-B outputs, or the repository’s exploratory personalization/memory code as formal evidence in this revision.

Summary of Revision Actions

1. **Scope narrowed to retrieval-grounded editable control.** The manuscript now centers on Texture Resonance Retrieval (TRR) as a retrieval representation for executable preset control. Personalization, AEM memory, and related extensions are explicitly presented as exploratory rather than validated core contributions.
2. **Objective claims upgraded to the P0 Protocol-A baseline matrix.** A post-hoc audit of the legacy song-name-based Protocol-A list found 16 cross-split shared resolved audio paths. We retain this audit for split transparency and now use the P0 204-query Protocol-A package as the main baseline matrix. The revised comparison includes TRR, Wav2Vec, FeatureNN, CLAP, PaSST, and PANNs on the shared 204-query / 1063-KB setup.
3. **Per-query diagnostics added.** We generated a derived analysis directly from the canonical Protocol-A per-query CSV for the stricter split, including win/loss/tie summaries, signed-delta diagnostics, and representative success/failure cases. These diagnostics are

now cited in the supplementary material so that the main result is not supported by mean values alone.

4. **TRR mechanism ablations added.** We added a controlled same-backbone ablation on the P0 split comparing Wav2Vec2 mean pooling, full unprojected Gram aggregation, single-layer projected Gram, projection dimensions, and with/without L2 normalization. The result supports a bounded mechanism claim: second-order Gram aggregation improves over same-backbone mean pooling and L2 normalization is important, while layer and projection choices remain meaningful hyperparameters.
5. **Stronger baselines and boundary baselines added.** We now include CLAP, PaSST, and PANNs in the P0 Protocol-A evidence chain. TRR remains strongest on normalized L2, Acc@0.1, Recall, Cosine, and Module among retrieval methods. We also add a diagnostic direct-regression boundary baseline using mean-pooled Wav2Vec2 features and a two-layer MLP trained only on the P0 knowledge base. Because that regressor predicts dense continuous parameter leaves rather than retrieved executable presets, we report it as a boundary condition rather than as a replacement for the retrieval leaderboard. Learned reranking remains future work.
6. **Protocol-C repositioned as diagnostic robustness evidence.** The older Protocol-C diagnostic used synthetic embedding-space degradation and remains only a boundary-condition artifact. The P0 Protocol-C result applies AWGN, MP3, reverb, and truncation to raw audio before TRR re-encoding, reuses the same 204-query split as E2, and compares fixed fusion against adaptive fusion. We additionally report current-code audio-only, pure-text, and parameter-oracle upper-bound rows as an execution-path audit. The revised manuscript no longer treats adaptive fusion as proof of a standalone main contribution or superiority over a pure-text boundary.
7. **Listening-study terminology and statistics corrected.** The study is consistently described as a multiple-stimulus listening test with a hidden reference and no explicit low-quality anchor. The revision reports the canonical nonparametric statistics already present in the repository: Friedman omnibus tests, Holm-corrected Wilcoxon signed-rank post-hoc tests, and rank-biserial effect sizes, together with an explicit limitation that the logs do not preserve device metadata, participant expertise, or the provenance/time budget of the manual baseline.
8. **Submission package synchronized.** The manuscript, supplement, response documents, and checklist were edited to remove stale claims about PaSST/PANNs, real-audio Protocol-C, near-duplicate placeholders, normalized metrics, and AEM learning curves.

Responses to the Main Reviewer Themes

1. Novelty and scope

We agree that the strongest supported contribution is not a full personalized agent system, but a retrieval-grounded parameter-control formulation centered on TRR. The revised manuscript therefore frames the contribution around texture-aware retrieval for editable preset control and avoids claims that would require additional evidence beyond the current benchmark.

2. Benchmark scale, split transparency, and statistical reporting

The objective results are now anchored to the shared P0 Protocol-A benchmark with $N_{\text{test}} = 204$ and $N_{\text{kb}} = 1063$. The revised manuscript and supplementary material report:

- the legacy split audit (16 shared resolved audio paths across split);
- the stricter anti-leakage split as audit context;
- the P0 baseline matrix including Wav2Vec, FeatureNN, CLAP, PaSST, and PANNs;
- method means including L2, Norm.L2, Acc@0.1, Recall, Cosine, and Module;
- Wilcoxon tests on per-query Norm.L2 for TRR versus each retrieval baseline;
- Holm-corrected decisions and effect-size caveats.

3. Stronger baselines

We now include *CLAP*, *PaSST*, and *PANNs* as stronger retrieval baselines in the P0 Protocol-A package. The revised manuscript therefore makes the narrower and better-supported statement that TRR is the strongest among the currently evaluated retrieval baselines on the shared P0 benchmark, including Wav2Vec, FeatureNN, CLAP, PaSST, and PANNs. We also include a diagnostic direct-regression MLP boundary baseline, but we do not claim to have completed an executable direct-parameter controller or learned reranking baseline.

3a. TRR mechanism evidence

We agree that the previous version did not sufficiently isolate why TRR works. We therefore added a controlled same-backbone mechanism ablation. The new supplementary table compares Wav2Vec2 mean pooling, full unprojected Gram aggregation, single-layer Gram variants for layers 4/5/6, projected multi-layer Gram variants with dimensions 16/32/64/128, and the same multi-layer setting without L2 normalization. The result supports a narrower mechanism claim: second-order Gram aggregation improves over same-backbone mean pooling, and L2 normalization is important. It also shows that the exact layer and projection choice matters, so the revised paper frames TRR as an effective texture-aware retrieval prior rather than a fully optimized universal representation.

4. Metric limitations

We also do *not* claim to have replaced the metric system with a normalized or categorical-aware suite. Instead, the revision clarifies the exact operational metric definitions and their limitations:

- flattened numeric leaves only;
- missing numeric keys treated as zero;
- no per-parameter normalization;
- within-protocol interpretation only.

These limitations are now stated explicitly in both the main paper and the supplementary material.

5. Protocol-C realism

We agree with the reviewers that synthetic embedding-space degradation is not equivalent to real audio corruption. The earlier diagnostic Protocol-C remains limited for that reason. The P0 Protocol-C package addresses this gap by applying real degradations to raw audio before re-encoding:

- AWGN at 20/10/5 dB;
- MP3 at 128/64/32 kbps;
- reverb with RT60 0.6/1.0;
- truncation to 50%/30%.

The retained claim is not that adaptive fusion dominates every condition or improves over a pure-text boundary. We added a boundary audit with audio-only, pure-text, and parameter-oracle upper-bound rows and revised the manuscript to treat Protocol-C as a diagnostic robustness analysis. The supported claim is narrower: audio-heavy fusion is fragile under realistic degradation, and retrieval stability depends strongly on modality weighting.

6. Listening-study rigor

The revised package now surfaces the canonical listening-study statistics already present in the repository:

- 26 participants and 910 ratings;
- Friedman omnibus tests for Trials 2–5 and Trials 6–10;
- Holm-corrected Wilcoxon signed-rank post-hoc tests;
- rank-biserial correlation (*rbc*);
- explicit statement that the raw HCAP label corresponds to the TRR-based system described in the paper.

We also state clearly that the TRR-based system and MusicGen are statistically indistinguishable in Trials 6–10 under the reported post-hoc test, so this block is interpreted as parity rather than superiority.

7. Scope mismatch between paper and repository

This concern was valid in earlier drafts. The revision addresses it by making the repository’s exploratory modules non-central:

- AEM / personalization is described as exploratory only;
- deprecated Protocol-B results are removed from the primary evidence path;
- legacy reports are no longer used as formal evidence;
- the main architecture figure now depicts only the verified retrieval-grounded pipeline.

Closing Statement

We appreciate the reviewers’ insistence on a tighter evidence chain. The revised submission is intentionally more conservative than earlier drafts: it makes fewer claims, but each retained claim is now tied to a canonical artifact with clearer statistical support and clearer scope boundaries.

Sincerely,
The Authors